

Beregningsteknik for Computer Ingeniører

MM 5: Complex numbers

Topics:

- Complex numbers (as a prerequisite to the following)
- Algebra and representations
- Complex roots
- Complex functions – derivatives and integrals

Rotation matrices

- Recall the (proper) rotation matrix from linear algebra which represents a counterclockwise rotation in the x-y plane

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad \text{where } 0 \leq \theta < 2\pi,$$

- When solving the eigenvalue problem

$$R(\theta)\vec{v} = \lambda\vec{v}.$$

- we find the characteristic equation

$$\lambda^2 - 2\lambda \cos \theta + 1 = 0, \quad \text{with roots} \quad \lambda = \cos \theta \pm \sqrt{\cos^2 \theta - 1}.$$

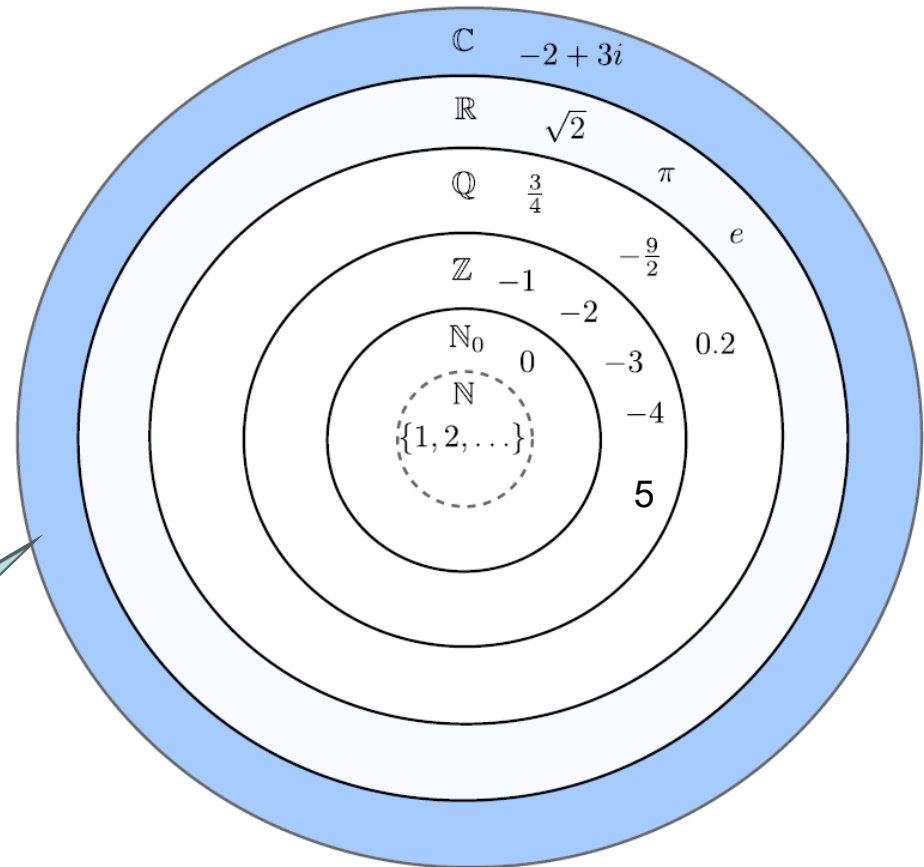
- ... no eigenvalues (when $(\theta \neq 0)_\pi$)!?

Complex numbers in Computer Engineering

1. **Signal Processing:** Complex numbers are essential in Fourier transforms, which are used to analyze and manipulate signals. This is crucial for applications like audio and image processing, where signals are often represented in the frequency domain.
2. **Control Systems:** In control theory, complex numbers help design stable and robust control systems. They are used to analyze system stability and response characteristics, which is vital for developing systems that can handle real-world challenges.
3. **Electromagnetism:** Complex numbers simplify the representation and calculation of electromagnetic fields and waves. This is particularly useful in designing and analyzing circuits and communication systems.
4. **Cryptography:** Complex numbers play a role in encryption algorithms, helping to secure communication by scrambling messages in a way that makes them unreadable to unauthorized users.
5. **Quantum Computing:** In quantum computing, complex numbers are fundamental. Quantum states are represented using complex probability amplitudes, and operations on these states involve complex number arithmetic.
6. **Computer Graphics:** Complex numbers are used in fractal generation and other graphical transformations. They help in creating intricate patterns and designs that are computationally intensive.

Numbers (number sets)

- Natural (+ incl. 0)
- Integers
- Rational
- Real
- Complex



Extension of real
numbers, $\mathbb{R} \subset \mathbb{C}$

C.F. Gauss
German
Mathematician
(1777-1855)

Complex numbers

- A **complex number** is an ordered set of real numbers $z = (x, y)$, where x is the **real part**, $\text{Re}()$, and y is the **imaginary part**, $\text{Im}()$, normally written as:
 - $z = x + iy$, (or for electrical engineers usually $z = x + jy$)
where $x = \text{Re}(z)$ and $y = \text{Im}(z)$
- Units are normally associated with physical quantities, but complex numbers have a "mathematical" unit, namely the **imaginary unit**

 $i = (0, 1) = 0 + i1$ – an (pure) imaginary number with the property that:
 $i^2 = -1$ or $i = \sqrt{-1}$ (definition)

"Complex" algebra

- Addition and subtraction

$$z = z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$$

$$z = z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2)$$

- Multiplication and division

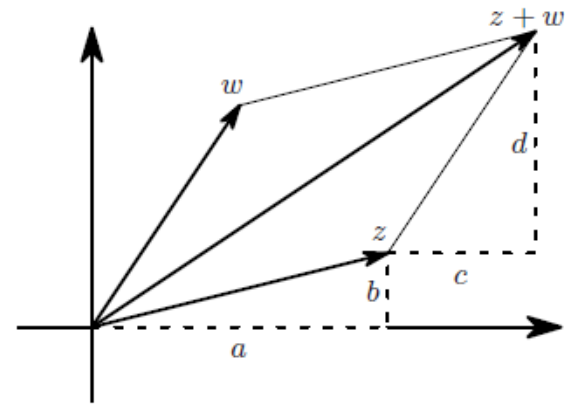
$$z = z_1 z_2 = (x_1 + iy_1)(x_2 + iy_2) = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1)$$

$$\begin{aligned} z = \frac{z_1}{z_2} &= \frac{x_1 + iy_1}{x_2 + iy_2} = \frac{(x_1 + iy_1)(x_2 - iy_2)}{(x_2 + iy_2)(x_2 - iy_2)}, z_2 \neq 0 \\ &= \frac{(x_1 + iy_1)(x_2 - iy_2)}{(x_2 + iy_2)(x_2 - iy_2)} = \frac{(x_1 x_2 + y_1 y_2)}{(x_2^2 + y_2^2)} + i \frac{(x_2 y_1 - x_1 y_2)}{(x_2^2 + y_2^2)} \end{aligned}$$

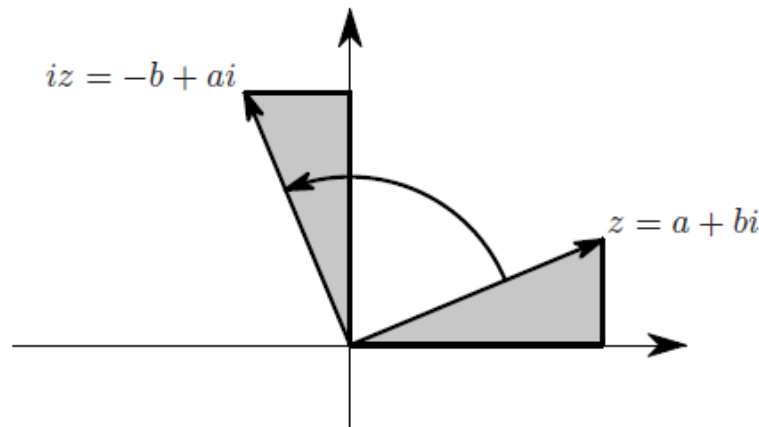
Complex (Argand) plane

- A complex number is much like a vector in two dimensions

- Addition/subtraction
($z = a + ib$; $w = c + id$)



- Multiplication by i
($z = a + ib$)



Complex conjugate

- The complex conjugate of $z = x + iy$ is

$$\bar{z} = z^* = x - iy \text{ (two common conventions denoting the conj.)}$$

- In fact, the ‘trick’ for complex division involves the conjugate

$$zz^* = x^2 + y^2$$

- Also

$$\operatorname{Re}(z) = x = \frac{1}{2}(z + z^*) \text{ and } \operatorname{Im}(z) = y = \frac{1}{2i}(z - z^*)$$

- As can be verified (try!)

$$\begin{aligned} (z^*)^* &= z; |z| = |z^*|; zz^* = |z|^2 & \left(\frac{z_1}{z_2}\right)^* &= \frac{z_1^*}{z_2^*} \\ (z_1 + z_2)^* &= z_1^* + z_2^*; (z_1 z_2)^* &= z_1^* z_2^* \end{aligned}$$

Polar form

- By change of variables $a = r\cos(\theta), b = r\sin(\theta)$

- we get the polar form of $z = a + ib$

$$z = r(\cos(\theta) + i\sin(\theta))$$

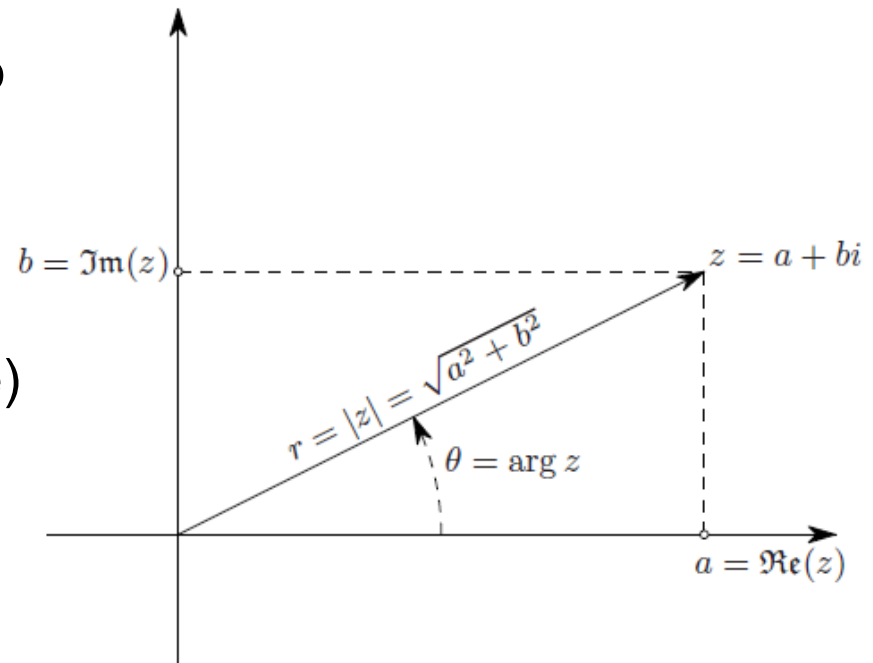
- with
 - The **modulus** (absolute value)

$$r = |z| = \sqrt{zz^*}$$

- And the **argument**

$$\theta = \arg(z) = \tan^{-1}(b/a)$$

- θ is defined from the positive x-axis
(**principal value** $\text{Arg}(z)$ in range $-\pi$ to $+\pi$)



$$z = r/\theta$$

Polar "algebra"

- The complex (multiplication) algebra can be done conveniently in polar form:

$$\begin{aligned} |z_1 z_2| &= |z_1| |z_2| \\ \arg(z_1 z_2) &= \arg(z_1) + \arg(z_2), \text{ mod } 2\pi! \end{aligned}$$

$$\arg(z^*) = -\arg(z)$$

$$\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$$

$$\begin{aligned} \arg(z_1/z_2) &= \arg(z_1) - \arg(z_2) \end{aligned}$$

- ... which we can then represent (back) in cartesian form (if so)
- For the complex (addition) algebra we have the "triangle inequality"

$$|z_1 + z_2| \leq |z_1| + |z_2| \quad (\text{with general extension})$$

"The absolute value of a sum cannot exceed the sum of the absolute values of the individual terms"

De Moivre's formula

- If we repeatedly multiply the same complex number and represent the result back in cartesian form we get (proof by induction!)

$$z^n = r^n(\cos(n\theta) + i\sin(n\theta)), n \in \mathbb{N}$$

(we could also multiply $z^{-1} = 1/z$ and extend to $n \in \mathbb{Z}$)

- For $r = 1$ we have De Moivre's formula

$$(\cos(\theta) + i(\sin(\theta)))^n = (\cos(n\theta) + i\sin(n\theta))$$

which is useful for representing powers of *cos* and *sin* (we require that real and imaginary parts equvalate) - EXERCISES

Roots

- We know from real roots that we can have ambiguity, e.g., $a^2 = 4$ has two roots, 2 and -2
- Although there is exactly one value z for each value of w in

$$z = w^n$$

$$w = \sqrt[n]{z}$$

there are precisely n distinct values of w – roots – for each z ! $z \neq 0$

- These follow from (using the generalized De Moivre)

$$w^n = R^n (\cos(n\phi) + i\sin(n\phi)) \quad w = R(\cos(\phi) + i\sin(\phi))$$

$$= z = r(\cos(\theta) + i\sin(\theta))$$

$$R^n = r \Leftrightarrow R = \sqrt[n]{r}$$

$$n\phi = \theta + 2k\pi \Leftrightarrow \phi = \theta/n + 2k\pi/n, k = \{0, 1, 2, \dots, n-1\}$$

- Taking the (principal) value of θ for z and $k = 0$ we get the principal value for the root $w = \sqrt[n]{z}$

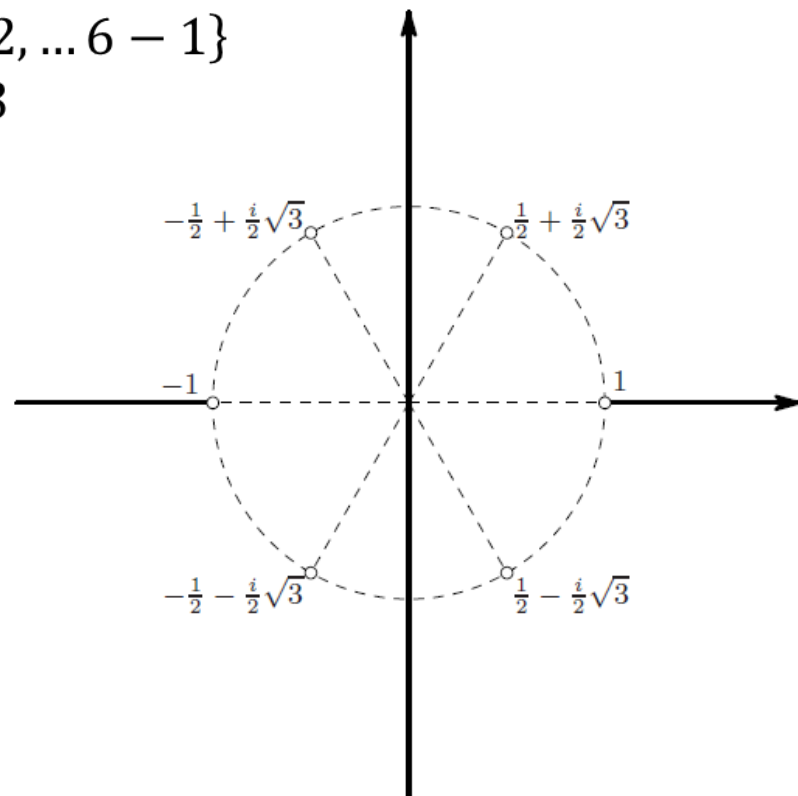
Roots

- Example: The six roots of $z = 1$ $w = \sqrt[6]{1} = R(\cos(\phi) + i\sin(\phi))$

$$\phi = 0/6 + 2k\pi/6, k = \{0, 1, 2, \dots, 6 - 1\}$$

... increments of $2\pi/6 = \pi/3$

$$R = \sqrt[6]{1}$$



Where's the principal root?